

Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal

PHYS DLA

Exercise Physiology

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Recommended Readings

- Rhoades, R. & Bell, D. (2012); Medical Physiology: Principles for Clinical Medicine, 6th Edition; Wolters Kluwer. Chapter 16 and 29
- Klabunde, Richard E. Cardiovascular physiology concepts, 3rd Edition, Chapter 7

Clinical cases:

Adapted from Cases and Problems in Physiology – Costanza

Objectives adapted from: The American Physiological Society-Medical Curriculum Objectives(2011), Cardiovascular

Practice Questions: BRS, Guyton, Pretest

Additional reading:

https://health.gov/sites/default/files/2019-10/DGA_Conversation-Starters.pdf



Learning Objectives and Outline

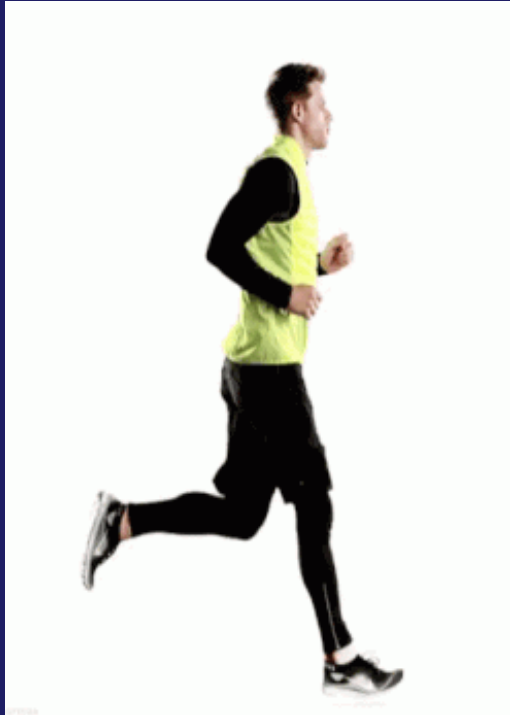
Learning Objective	Lecture Outline
SOM. MK.II.BPM1.3.CPR.2.PHYS.0186 : Describe the cardiac output, circulation, respiration response to physical exercise.	Definitions
SOM. MK.II.BPM1.3.CPR.2.PHYS.0187 : Explain the adaptations to training in the cardiovascular, respiratory, musculoskeletal systems and metabolic adaptations	Response to exercise
SOM.MK.II.BPM1.3.CPR.2.PHYS.0155 : Describe the cardiovascular consequences of exercise on peripheral resistance, and cardiac output	Response to exercise
SOM.MK.II.BPM1.3.CPR.2.PHYS.0156 : Contrast the neural and local control of skeletal muscle blood flow at rest and during exercise.	Response to exercise
SOM.MK.II.BPM1.3.CPR.2.PHYS.0157 : Discuss the interaction of neural and humoral control mechanisms and contrast their relative dominance in the skeletal muscle vascular beds.	Response to exercise
SOM. MK.II.BPM1.3.CPR.2.PHYS.0188 : Discuss the benefits of regular physical activity and exercise.	Response to training
SOM. MK.II.BPM1.3.CPR.2.PHYS.0189 : Discuss the risks of exercise	Benefits of exercise
SOM. MK.II.BPM1.3.CPR.2.PHYS.0190 : Describe the medical evaluations necessary prior to exercise	Risks of exercise
SOM. MK.II.BPM1.3.CPR.2.PHYS.0337 : Explain the importance of health education and promotion in health care as well as within culturally diverse populations.	Exercise prescription
SOM. MK.II.BPM1.3.CPR.2.PHYS.0338 : Demonstrate appropriate techniques for patient education regarding exercise recommendations based on WHO guidelines	Patient education



What is exercise?



Definitions



Physical activity is NOT exercise.

Physical activity is defined as bodily movement produced by the contraction of skeletal muscle that increases energy expenditure above the basal level – WHO

Categories of physical activity include - household, work, leisure activity, and transportation.

Exercise is defined as bodily activity (usually planned, structured, repetitive and purposeful), which enhances or maintains physical fitness and overall health or wellness.

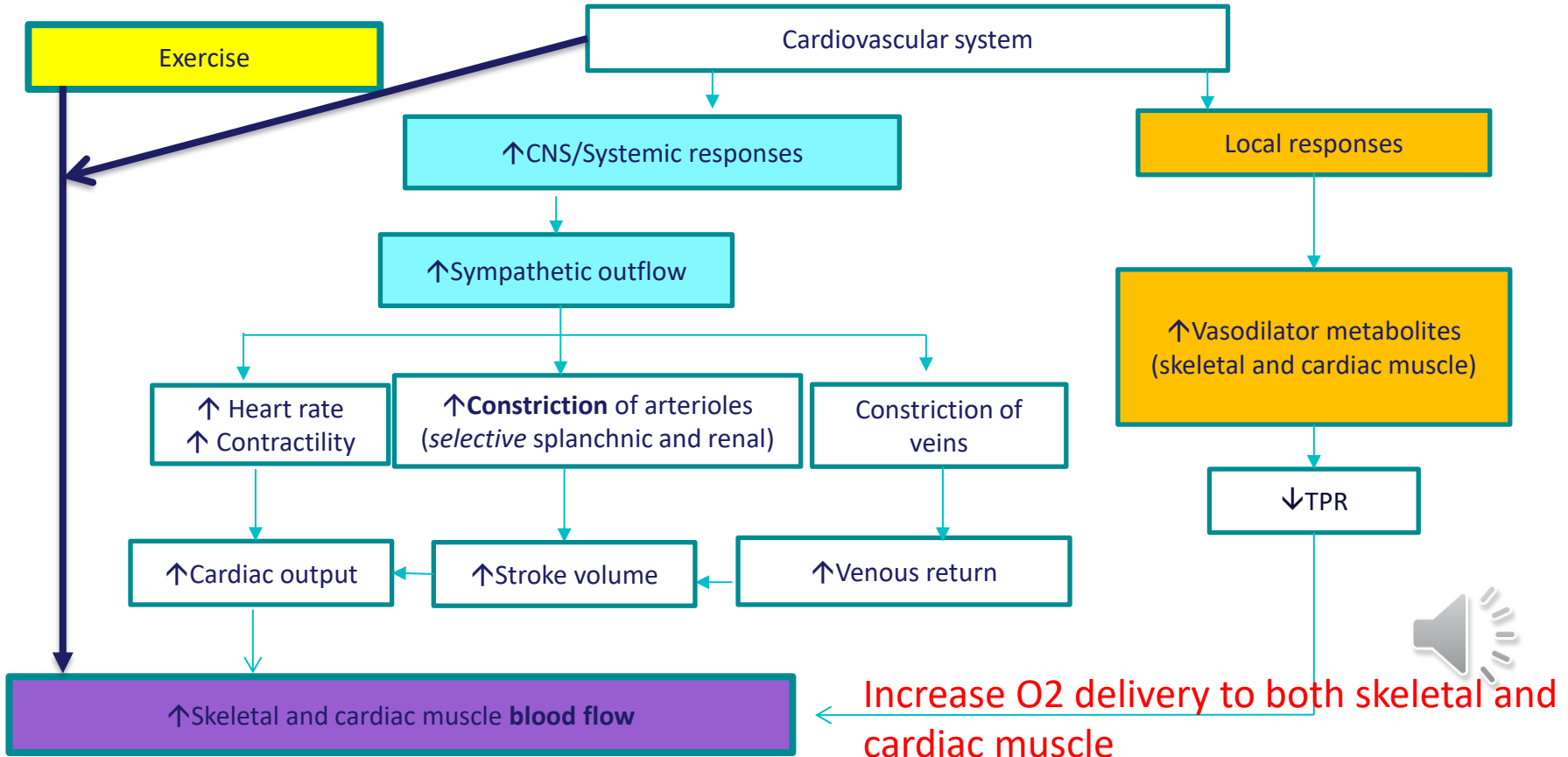
Exercise requires efficient supply to skeletal muscles, via the integration of the cardiovascular, circulatory and respiratory systems.



What is the goal of the
CARDIOVASCULAR
systemic response to moderate
exercise?



The cardiovascular system response to exercise



Skeletal Muscle Blood Flow: At rest & during Exercise

At Rest:

- Skeletal Muscles receive about 20% of the cardiac output.
- Tonic, sympathetic-mediated vasoconstriction dominates blood flow.
- This basal vasoconstrictor tone also contributes to the normal total peripheral resistance (TPR) and mean arterial pressure (MAP).
- Vasoactive metabolites are very low and do not override sympathetic control.

During Exercise:

- Vasoactive metabolites are released by active and reactive hyperemia and override sympathetic vasoconstriction.
- Major vasodilators include lactate, K^+ , CO_2 , and adenosine.
- Blood flow to skeletal muscles may increase up to 80% of the cardiac output.
- Changes in TPR and MAP depend on the type of exercise (static/dynamic).



Phasic versus Static muscle contraction during exercise

Phasic or Rhythmic muscle contraction:

e.g. Exercises like running, swimming.

Muscle blood flow increases by **active hyperemia** due to metabolites (see fig).

Generalized sympathetic stimulation maintains TPR and MAP.

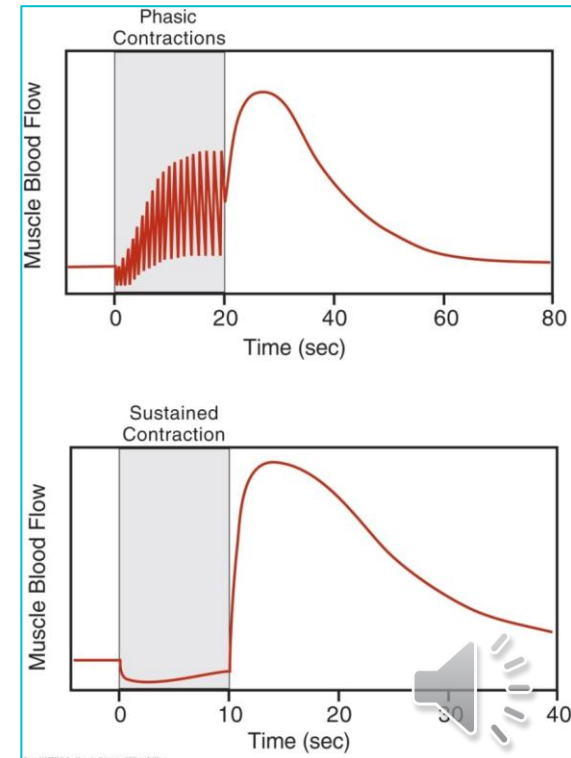
Static muscle contraction:

e.g. Exercise like weightlifting.

Sustained contractions block blood flow and increase local resistance.

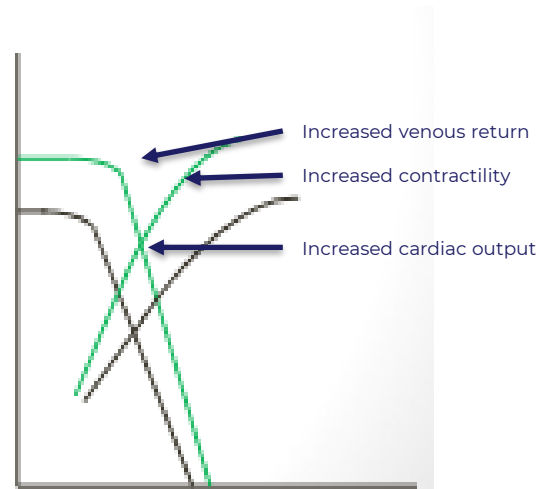
Increased TPR and cardiac output can lead to increased MAP.

Muscle blood flow increases after stoppage of contraction by **reactive hyperemia** due to build-up of metabolites (see fig).



The cardiovascular system increases blood flow to both skeletal and cardiac muscles in response to exercise

PARAMETER	RESPONSE TO EXERCISE
Heart rate (and contractility)	↑↑ (sympathetic stimulation)
Stroke volume	↑
Pulse pressure	↑ (increased stroke volume)
Cardiac output	↑↑
Venous return	↑
Mean arterial pressure	↑ (slight)
Total peripheral resistance (TPR)	↓↓ (metabolites cause vasodilation in skeletal muscle)
Arteriovenous O ₂ difference	↑↑ (increased O ₂ consumption in tissues)

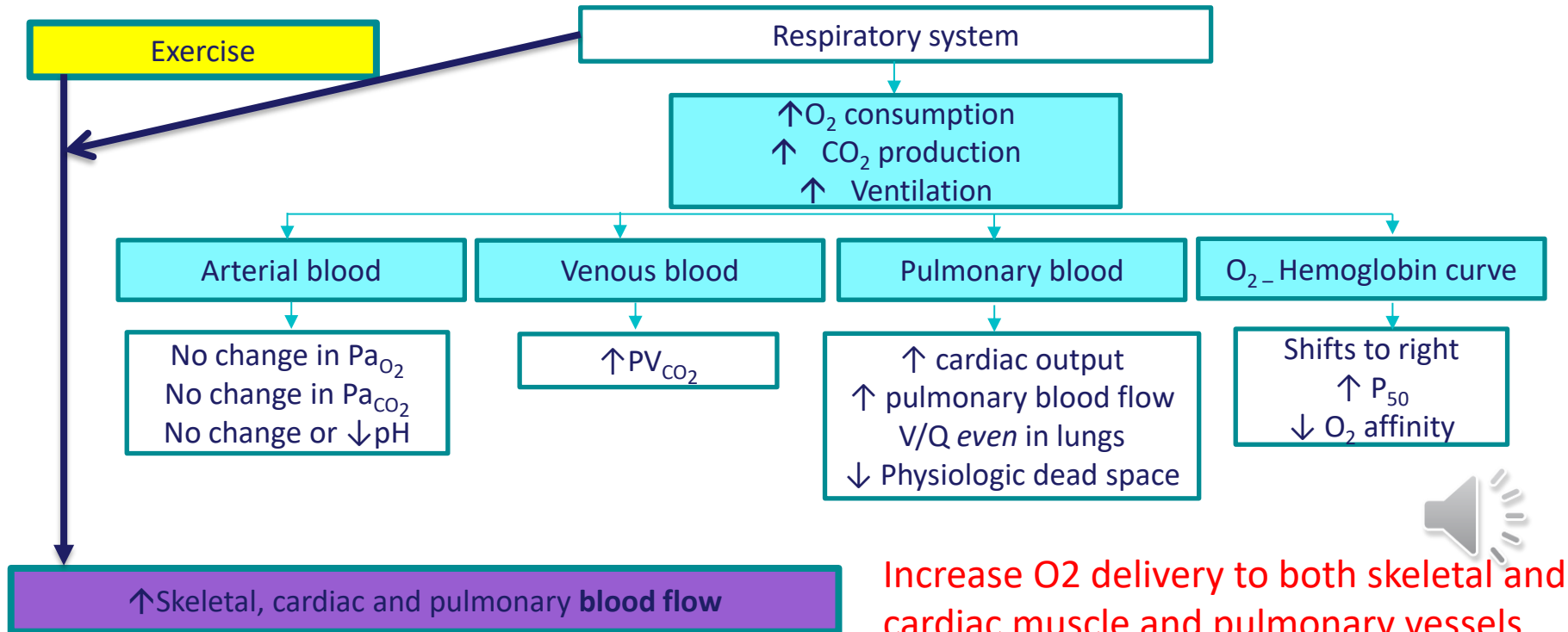


Increase O₂ delivery to both skeletal and cardiac muscle

What is the goal of the
RESPIRATORY
systemic response to moderate
exercise?

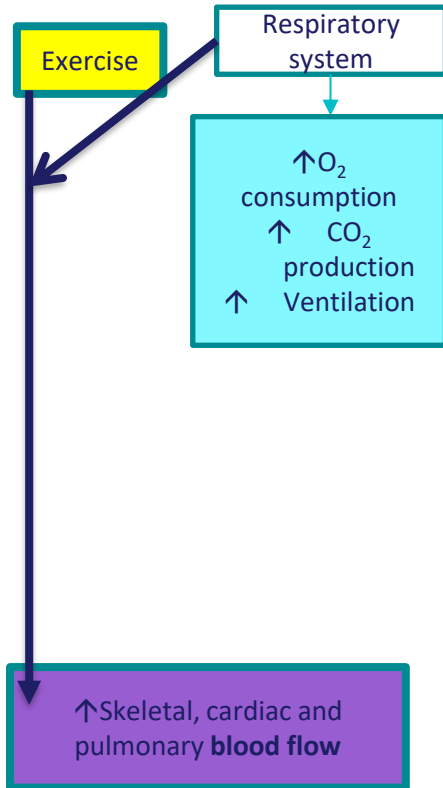


The respiratory system response to exercise

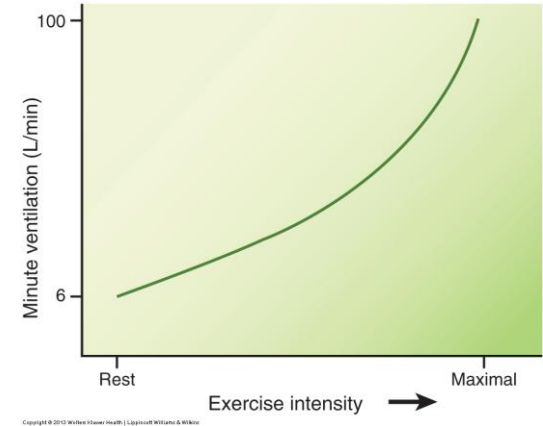


Increase O_2 delivery to both skeletal and cardiac muscle and pulmonary vessels

Respiratory system increases oxygen supply to the demanding skeletal muscles



- As **O₂ consumption increases** due to the exercising muscles, the ventilation rate will increase to supply the O₂.
- Muscle and joint receptors provide immediate feedback to medullary inspiratory center to increase the ventilatory rate.
- Exercising muscles extract the O₂ and **adds CO₂** to venous blood. This also causes the ventilation rate to increase to 'get rid' of this extra CO₂.
- Overall **ventilation rate increases** up to approximately 10-fold with intense exertion.

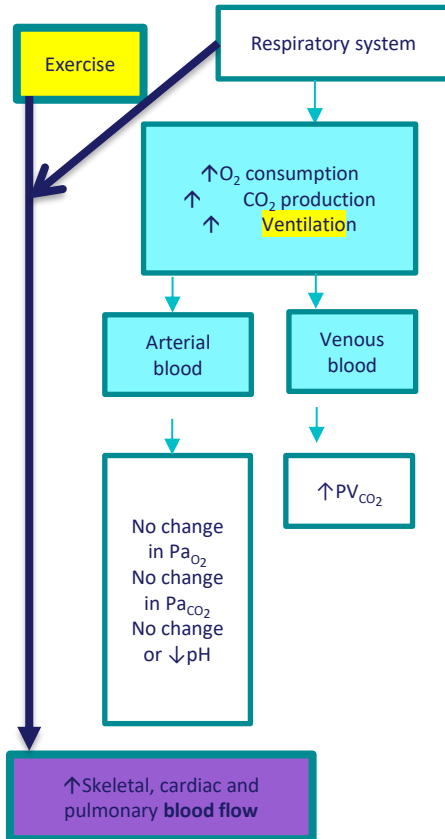


Respiratory muscles accomplish increase in ventilation primarily by increasing tidal volume,



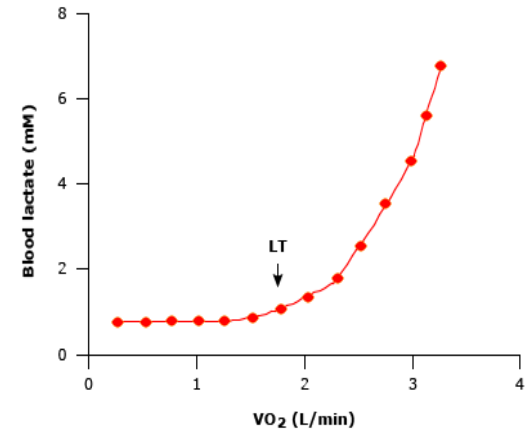
Increase O₂ delivery to both skeletal and cardiac muscle and pulmonary vessels

Respiratory system increases oxygen supply to the demanding skeletal muscles



- The increase in ventilation rate and efficient gas exchange maintains homeostasis.
- Mean arterial PO_2 and PCO_2 do not change during exercise.
- Arterial pH remains normal under moderate exercise.
- However, under strenuous exercise, lactate builds up when it is produced faster than it can be metabolised via the TCA cycle – anaerobic threshold or lactate threshold
- Increase in lactic acid also increases the ventilation rate.

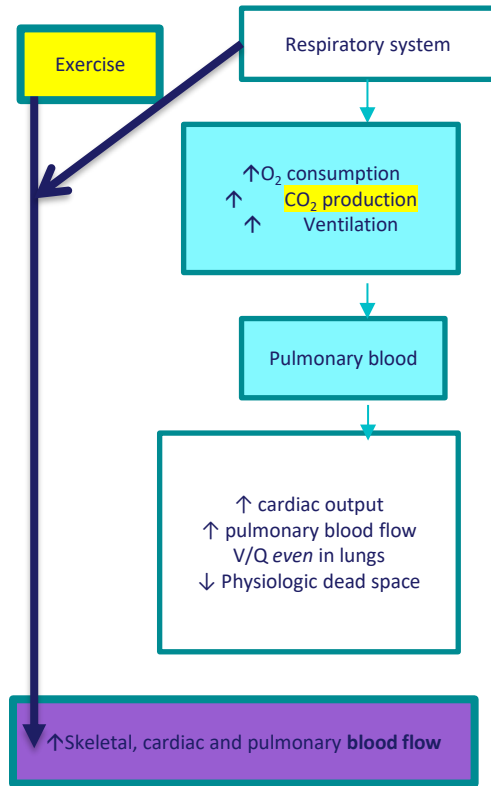
Lactate threshold during exercise



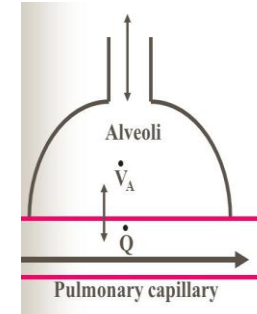
Schematic representation of the normal blood lactate - oxygen uptake (VO_2) relationship during incremental exercise. The lactate threshold (LT) denotes the onset of a sustained rise in blood lactate.



Respiratory system increases oxygen supply to the demanding skeletal muscles



- Cardiac output increases during exercise to meet the tissue demand for O_2 .
- Pulmonary blood flow increases because of the increased cardiac output and decreased pulmonary resistance results in increased pulmonary capillary perfusion.
- Gas exchange becomes more efficient.
- V/Q remains evenly distributed –because both perfusion and ventilation has increased.



- **Ventilation – perfusion ration**
 - Expressed as V_A/Q
 - Where
 - V_A = alveolar ventilation for a given alveolus
 - Q = capillary blood flow for the same alveolus

The match between air in the alveoli (due to ventilation) and blood in the capillary (due to perfusion) is equal in all regions.



Increase O_2 delivery to both skeletal and cardiac muscle and pulmonary vessels

Respiratory system increases oxygen supply to the demanding skeletal muscles

With increased blood lactate,
Arterial pH may decrease slightly

The oxygen dissociation curve shifts to the right because of:

drop in pH,
increase in PCO_2
increase in basal metabolic rate

A right-ward shift means:

Decrease affinity of hemoglobin for oxygen
Easier to offload O_2 to skeletal muscle

Oxygen-hemoglobin Dissociation: Exercise

Factors shifting curve to **right**

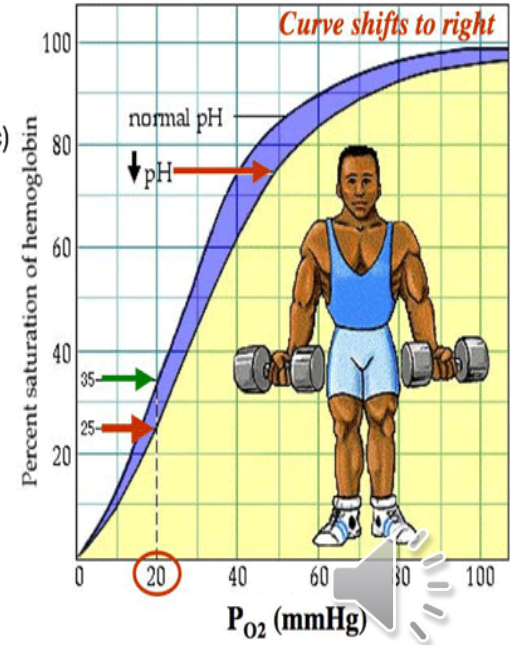
↓ pH (more acidic)

↑ Temperature

↑ P_{CO_2}

↑ 2,3-BPG

Called
Bohr effect



How does training alter the response of the systems?



How does training alter the response of the systems?

Most studies show that cardiovascular training requires a minimum of 30 minutes, 3 times a week to guarantee increased aerobic capacity in about 8 to 12 weeks.

Training increases the efficiency of the cardio-pulmonary system in providing oxygen and nutrient rich blood to skeletal and cardiac muscles for ATP production and removal of waste products from those muscles.

Cardiovascular

Muscle mass of ventricles increases
(usually symmetrical hypertrophy)

Increase the force of contraction

LV size generally normal or increased but
normal LV systolic function and no
regional wall abnormalities

Increase the force of contraction

Increased capillary capacity

Decrease TPR

Respiratory

Lactate levels at VO_2 max for a given level of work less in trained subjects than sedentary controls

Lactate threshold at above 40% of the predicted VO_2 Contrast at a lower % in exercise among patients with cardiovascular disease, higher in endurance

Decrease in minute ventilation necessary to achieve a given VO_2 . Lower minute ventilation a function of ventilatory training rather than changes in control of respiration athletes

Lung diffusing capacity, lung mechanics and lung volumes change little with training

Training doesn't improve the vital capacity. Even exercise designed specifically to increase inspiratory muscle strength elevates vital capacity by only 3%.

The primary respiratory changes with training are as a result of lower lactate production which reduces ventilatory demands at heavy work levels



How does training alter the response of the systems?

Other adaptations

Musculoskeletal system

- Type of training affects the muscular adaptations e.g.
 - Endurance training – mitochondrial biogenesis, fast to slow fibre transformation, expansion of muscle capillary bed, changes in substrate metabolism and increased cardiac output
 - Resistance training- increase size protein content of muscle fibres

Metabolic adaptation to endurance training

- Increase in size and number of muscle mitochondria. Increased TCA intermediates in muscle mitochondria
- Increase storage of glycogen in skeletal muscle
- Increased fat utilisation, spares glycogen stores

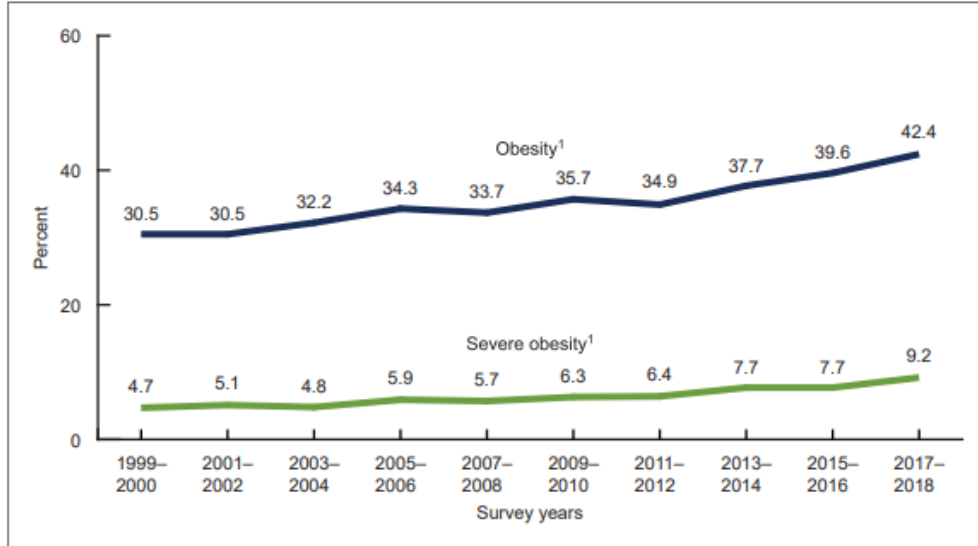


Why should a physician prescribe exercise?



Why should physicians prescribe exercise?

Figure 4. Trends in age-adjusted obesity and severe obesity prevalence among adults aged 20 and over: United States, 1999–2000 through 2017–2018



https://www.ncbi.nlm.nih.gov/books/NBK278973/bin/clin-prblms-obesity_figure1-1024x699.jpg



Why should physicians prescribe exercise?

Evidence for health benefits associated with regular physical activity (2008 Physical Activity Guideline)

Adults and older adults

Strong evidence

- Lower risk of early death
- Lower risk of coronary heart disease
- Lower risk of stroke
- Lower risk of high blood pressure
- Lower risk of adverse blood lipid profile
- Lower risk of type 2 diabetes
- Lower risk of metabolic syndrome
- Lower risk of colon cancer
- Lower risk of breast cancer
- Prevention of weight gain
- Weight loss, particularly when combined with reduced calorie intake
- Improved cardiorespiratory and muscular fitness
- Prevention of falls
- Reduced depression
- Better cognitive function (for older adults)

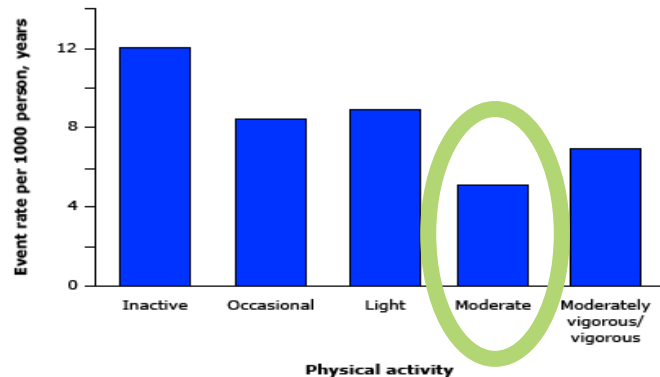
Moderate to strong evidence

- Better functional health (for older adults)
- Reduced abdominal obesity

Moderate evidence

- Lower risk of hip fracture
- Lower risk of lung cancer
- Lower risk of endometrial cancer
- Weight maintenance after weight loss
- Increased bone density
- Improved sleep quality

Beneficial effects of any physical activity on coronary heart disease



Coronary events are less frequent among those who exercise. In a study of 5159 men, aged 40 to 49 years, followed for an average of almost 19 years, the age-adjusted coronary heart disease event rate per 1000 person-years is lower in those who perform any physical activity compared to inactive subjects.

Data from: Wannamethee SG, Shaper AG, Alberti KG. Arch Intern Med 2000; 160:2108.



WHO Exercise Recommendations

Adults aged 18–64 years should:

- Do at least 150–300 minutes of moderate-intensity aerobic physical activity;
- Do at least 75–150 minutes of vigorous-intensity aerobic physical activity
- Limit the amount of time spent being sedentary.
- Muscle-strengthening activities for all major muscle groups on 2 or more days a week has been shown to provide added benefit

Benefits include:

- improve bone and functional health;
- reduce the risk of hypertension, coronary heart disease, stroke, diabetes, various types of cancer (including breast cancer and colon cancer)
- improve mental health (reduced symptoms of anxiety and depression), cognitive health and sleep



What are the risks associated with exercise?



Are there risks associated with exercise?

For the majority **benefits far outweigh possible risks**

- Musculoskeletal injury most common risk
- Increased risk of arrhythmia during exercise in patients with previous rhythm disturbances and heart disease
- Sudden cardiac death.
 - Patients < 35 years more likely due to underlying myocarditis, hypertrophic cardiomyopathy, coronary anomalies;
 - Patients >35 years most likely due to coronary artery disease
- Rhabdomyolysis - a syndrome caused by injury to skeletal muscle and involves leakage of large quantities of potentially toxic intracellular contents into plasma
- Exercise induced bronchoconstriction – asthmatic presentation
- Hyper/hypothermia, dehydration
- In intense training - amenorrhoea, infertility especially in women with low body weight.



Medical evaluation prior to the start of exercise is important

What are we assessing?

- Appropriate history with age, past medical history, focusing especially on cardiac, pulmonary, orthopaedic history.
- History of diabetes, hypertension, hyperlipidaemia.
- General physical condition
- Medication use e.g. Sulphonylureas, insulin, betablockers, nitrates
- Anticipated type of exercise
- Disabilities
- Perform periodic exercise stress testing in asymptomatic clients with multiple cardiovascular risk factors e.g. Hypertension, diabetes, hypercholesterolaemia, smoking.
- Screening not necessary for asymptomatic clients at low risk of coronary artery disease.
- H/O premature myocardial infarct or sudden cardiac death in a first degree relative < 60 yrs old.
- Appropriate screening blood tests based on history.



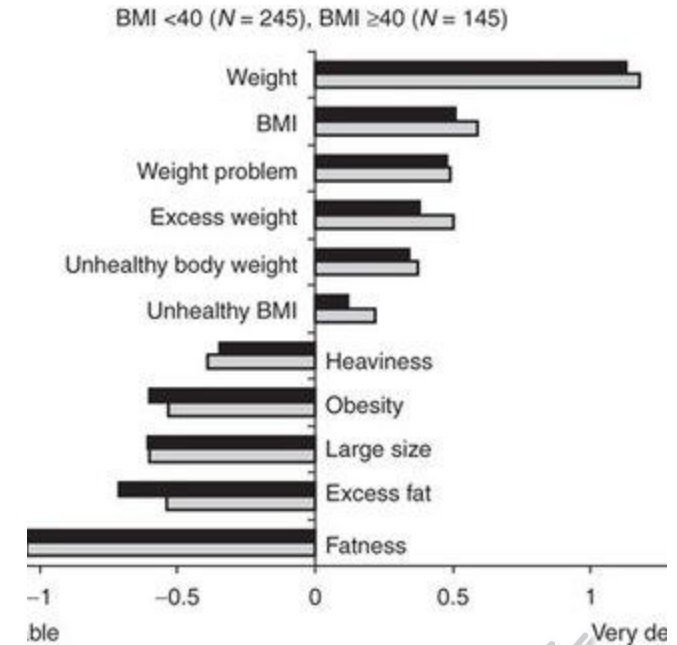
Using the exercise prescription for patient education



The exercise prescription – patient education tips

Before initiating diet and exercise counselling, mention the health risks associated with overweight and obesity. Ask patients if you can talk with them about their general health, including weight.

Research has shown that patients prefer the terms “weight” (first) and “BMI” (second) when talking about obesity, and dislike the terms “fatness,” “excess fat,” and “obesity.” Preferred and non-preferred terms generally were the same across social and economic factors. It is important to use non-judgmental and inclusive language when providing patient education to ensure an appropriate doctor-patient relationship of trust and support



The exercise prescription – patient education

No one exercise ‘fits all’.

The following four principles of exercise guide recommendations:

1. **Frequency – The number of days per week dedicated to the activity.** Activity should be enjoyable, accessible, within financial means, safe for the individual, at least three times per week
2. **Intensity – How hard a person works to do the activity.** Moderate intensity exercise
3. **Time –The length of time in which an activity is performed.** Duration for at least 50 minutes.
4. **Type – the Mode of exercise performed.** Stress that even modest increases in physical activity are associated with improved health outcomes

It is also vital to counsel patients on healthy diet habits. Open the discussion about weight in a respectful and nonjudgmental way. Patients may be more open if they feel respected



A 33-year-old woman comes to the clinic for a routine evaluation. She is in good health. Her blood pressure 120/76 mmHg, pulse is 60/min, respirations are 20/min. She completes an exercise stress test on a treadmill. Which of the following cardiovascular changes is most likely present in this patient?

A. A	Contractility	Venous return	Systemic vascular resistance
B. B	Decreased	Increased	Decreased
C. C	Increased	Increased	Decreased
D. D	Increased	Increased	Increased
E. E	Decreased	Decreased	Decreased
	Increased	Unchanged	Unchanged



A 33-year-old woman comes to the clinic for a preconception evaluation. She is in good health with regular menstrual cycles. She is a professional cyclist with no known uses of drugs or alcohol. Vital signs are blood pressure 114/66 mmHg and pulse is 60/min. Compared to a person of similar age and sex with sedentary lifestyle, which of the following cardiovascular and respiratory changes is most likely present in this patient?

	LV muscle mass	LV systolic function	Lactate levels at $\dot{V}O_2$
A. A			
B. B	Decreased	Increased	Increased
C. C	Increased	Unchanged	Decreased
D. D	Unchanged	Increased	Increased
E. E	Increased	Increased	Increased
	Decreased	Unchanged	Unchanged



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Increased Cardiac Contractility: Enhanced by sympathetic stimulation, leading to a higher stroke volume and cardiac output.

Increased Venous Return: Facilitated by the muscle pump, respiratory pump, and venoconstriction, which increases preload and stroke volume.

Decreased Systemic Vascular Resistance: Due to vasodilation in the arterioles of active muscles, allowing for increased blood flow to meet metabolic demands.

2. A 33-year-old woman comes to the clinic for a preconception evaluation. She is in good health with regular menstrual cycles. She is a professional cyclist with no known uses of drugs or alcohol. Vital signs are blood pressure 114/66 mmHg and pulse is 60/min. Compared to a person of similar age and sex with sedentary lifestyle, which of the following cardiovascular and respiratory changes is most likely present in this patient?

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D	Increased	Decreased	Increased
E	Decreased	Unchanged	Unchanged

Increased Left Ventricular Muscle Mass: Adaptations such as eccentric or concentric hypertrophy allow for greater stroke volume and cardiac output, essential for sustaining high levels of physical activity.

Unchanged or Enhanced Left Ventricular Systolic Function: The heart's ability to contract and pump blood is either maintained or improved, particularly during exercise, leading to more efficient cardiovascular performance.

Optimized Lactate Handling: Athletes have lower lactate levels at submaximal exercise intensities and an increased lactate threshold, allowing them to perform at higher intensities with less fatigue.

WARNING

EXERCISE
HAS BEEN KNOWN
TO CAUSE HEALTH
& HAPPINESS

